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Quantitative decision science

M.A. Actuarial Economics · Madras School of Economics · Class of 2028
Summer 2027 internships · Seven core projects with source-linked evidence

Operational forecasting, causal decisions and bank-risk validation lead the portfolio. Finance, economics, treasury and insurance add distinct role-specific evidence. Each project makes its data, methods, results and limitations inspectable.

1. Operational forecasting and allocation

FreshRetailNet-LT observed sales support time-aware forecasts and capacity-constrained allocation scenarios. The implementation includes a typed ingestion ledger, immutable input snapshots and asynchronous jobs. Stockout-adjusted forecasts and operating effects remain scenarios, not observed retailer savings.

Operations analytics, forecasting, decision systems
Code and research · Evidence browser

2. Causal customer decisions

A 64,000-customer randomized campaign did not establish that personalized targeting beats the selected simple policy. A separate 873,520-row Criteo cohort tests released-sample response and selection sensitivity; it does not identify original-population causal profit.

Product analytics, experimentation, causal inference
Code and research · Evidence browser

3. Bank-risk model validation

237,253 annual bank observations support a purged temporal failure-monitoring study with separate calibration and policy stages. Later monitoring missed 9 of 10 failures, restricting use despite stronger earlier ranking. Bank failure is not borrower default or regulatory approval.

Bank risk, model validation, risk analytics
Code and research · Evidence browser

4. Corporate Finance Decision Lab

Nine issuer reports, a traced financial spread and a funded-growth policy model. Final extraction: 33 supported correct /36 financial targets for both methods, three justified abstentions. 519 independent valuation checks and a reconciled formula workbook.

Corporate finance, FP&A, financial research and finance data
Code and research · Evidence browser

Research portfolio

5. Indian inflation release vintages

48 official releases and 144 vintage observations. An 18-release final check retains persistence: RMSE 0.8185 pp versus 1.6901 for fixed AR(1). Release dates and the later base revision remain separate. No current inflation forecast is claimed.

Economic research, time series, publication-aware data
Code and research · Evidence browser

6. Treasury ALM and liquidity

Interest-rate hedging, published Treasury curves and hypothetical benefit cash flows. Four hedge policies, 56 rate stresses and 108 liquidity-runoff cases. Independent Decimal and formula-workbook reconciliations support the modelling.

Treasury, ALM, financial risk
Code and research · Evidence browser

7. Insurance pricing and reserve uncertainty

French motor pricing and a separate US auto-reserving study. Fixed pricing comparisons and reserve interval scores expose conditional-calibration and coverage failures. The two populations are not joined.

Actuarial analytics, insurance risk, uncertainty modelling
Code and research · Evidence browser

Choose a starting point

Bank/model risk: studies 3 and 6. Insurance: 7 and 6. Corporate finance/FP&A: 4 and 5. Product/business analytics: 2 and 1. Economic research: 5 and 2. The shared standard is traceable source data, explicit information timing, appropriate uncertainty and a decision that follows the evidence.

Open the seven-study portfolio · GitHub · LinkedIn